

Is $\sum a_n$ is C or D

① $\lim_{n \rightarrow \infty} S_n = L \Rightarrow \sum a_n = C$
 $= \pm \infty \Rightarrow \sum a_n = D$

② T.S $\sum \omega - \omega$	G.S $\sum ar^{n-1}$ $r < 1 \Rightarrow C$ $r > 1 \Rightarrow D$	P.S 11.2 $p > 1 \Rightarrow C$ $p < 1 \Rightarrow D$
A.S $\frac{1}{n} = D$		

③ Tests:

- ☒ D.T
- ☒ I.t
- ☒ B.C.t
- ☒ L.C.t
- ☒ A.S.t
- ☒ rative.t
- ☒ root.t
- ☒ A.C or C.C

☒ Powe.s 11.8 + 11.9

☒ M.g, T.S

$$\textcircled{1} \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

$r=1$ I.C. = $(-1, 1)$, $|x| < 1$

$$\textcircled{2} \tan^{-1} x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)}$$

$r=1$ I.C. = $[-1, 1]$, $|x| \leq 1$

$$\textcircled{3} \ln(1+x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1}$$

$r=1$ I.C. = $(-1, 1)$, $|x| < 1$

$$\textcircled{4} e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$r=\infty$ I.C. = $(-\infty, \infty)$

$$\textcircled{5} \cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

$r=\infty$ I.C. = $(-\infty, \infty)$

$$\textcircled{6} \sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

$r=\infty$ I.C. = $(-\infty, \infty)$

$$\textcircled{7} \text{Binomial's}$$

$$(1+x)^a = 1 + \sum_{n=0}^{\infty} \frac{a(a-1)(a-2)\dots(a-n+1)}{n!} x^n$$

$r=1$ I.C. = $(-1, 1)$, $|x| < 1$

$$r = \frac{b-a}{2}$$

$$\textcircled{8} \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

$$\textcircled{9} \lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$$

$$\textcircled{10} \lim_{n \rightarrow \infty} \frac{n^n}{n!} = \infty$$

$$= \ln(n) \ll n^a \ll a^n \ll n! \ll n^n$$

$$\textcircled{11} \lim_{n \rightarrow \infty} \frac{e^n}{n!} \text{ or } \frac{n!}{e^n}$$

$\downarrow$ $\downarrow$
0 ∞

$$\textcircled{12} \pi/4 \leq \tan^{-1}(n) \leq \pi/2$$

$$0 \leq \frac{\sin^2(n)}{\cos^2(n)} \leq 1$$

$$-1 \leq \frac{\sin(n)}{\cos(n)} \leq 1$$

$$0 \leq \tanh(n) \leq 1$$

$$\textcircled{13} \frac{\cos^2 x}{\cosh^2 x} = \frac{1}{2} \left[\frac{1 + \cos(2x)}{\cosh(2x)} \right]$$

$$\textcircled{14} (1+x)^{K^4} = \sum (u^K) u^n$$

$$\textcircled{15} \lim_{n \rightarrow \infty} \frac{\sin(1/n)}{1/n} = 1$$

$$\textcircled{16} \lim_{n \rightarrow \infty} \frac{\tan(1/n)}{1/n} = 1$$

$$\textcircled{17} \lim_{n \rightarrow \infty} \left(1 \pm \frac{\#}{n} \right)^{bn} = e^{b\#}$$

$$\textcircled{18} \lim_{n \rightarrow A} \left(1 \pm \frac{a}{bn+c} \right)^{\#n} = e^{\pm \frac{a\#}{b}}$$

$$\textcircled{19} (-1)^n = \cos(n\pi)$$

① Telescoping. S:-

$$\sum \text{نوع} - \text{نوع}$$

① $a_n =$ الـ

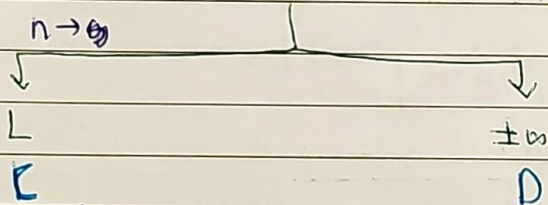
② $S_n = a_1 + a_2 + a_3 + \dots + a_{n-1} + a_n$

اخرى بدلان $n \rightarrow n-1$ $r, c, 1$

③ $\sum$ تطبق

④ $\sum_{n \rightarrow \infty} S_n =$

∞



$\therefore \sum a_n$ is

② Geometric. S:-

$$\sum_{n=0}^{\infty} a r^{n-1}$$

① a_1 and a_2 $|r| < 1$ $|r| > 1$

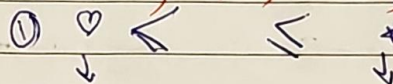
② $r = \frac{a_2}{a_1}$ C D

③ $\sum a_n$ is

④ $S = \frac{a}{1-r}$, $S_n = S_1 - S_2$

③ B.C.T :-

$\tan^{-1}$, $\sin$, $\cos$, "h"



$\sum \rho$

$\sum G.S$

$|r| < 1$ $|r| > 1$

C $\leftarrow$ C

D $\rightarrow$ D

④ L.C.T :-

① $\sum \frac{\text{اخرى}}{\text{اخرى}} = \frac{n}{n}$

② $\sum \frac{1}{\sin(\dots)} = \frac{\text{اخرى}}{\text{اخرى}} \left(\frac{\text{زاوية}}{\text{اخرى}} \right) b_n$

③ $\sum \frac{\text{اخرى}}{\text{اخرى}} = \frac{n}{n} \text{ (e)}$

④ $\sum \frac{\text{اخرى}}{\text{اخرى}} = \frac{\text{اخرى}}{\text{اخرى}}$

① $b_n =$

② $\sum b_n = \sum p$ or G.S

③ $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \text{اخرى} \cdot \text{اخرى} b_n > 0$

$\therefore \sum a_n$ is

5 A.S.T:

$$\sum (-1)^n b_n$$

① $\lim_{n \rightarrow \infty} b_n = 0$, $\lim_{n \rightarrow \infty} b_n = \infty$ [D]

$0 < b_n < \frac{1}{n}$ or $\frac{1}{n}$

② $b_n \downarrow$

$f'(x) < 0$

$\exists \epsilon$

< 0

$b_{n+1} < b_n$

b_1, b_2, b_3
 $\downarrow \downarrow \downarrow$

$\therefore \sum (-1)^n b_n$ is [C]

6 A.C, C.C, div:

D

C

A.8

① $\lim_{n \rightarrow \infty} |a_n| = 0$

$\lim_{n \rightarrow \infty} |a_n| \neq 0$
D

② $b_n \downarrow$

C.C

① $\sum |a_n|$

② $\sum |a_n| =$ L.C.t, B.C.t, P, G.S

③ C D
A.C

④ $\lim_{n \rightarrow \infty} |a_n| \neq 0 = 0$
D

⑤ $b_n \downarrow$
C.C

7 D.T:

① $\sum a_n$

② $\lim_{n \rightarrow \infty} a_n = n > 0, \therefore \sum$

L [D]

$= 0$ X

8 ratio.T:

$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \frac{n!}{n^p, p^n, n^n}$

① $a_n = a_{n+1} =$

② $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = a_{n+1} - a_n$

③ $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$ or > 1 or $= 1$

< 1

> 1 or ∞

$= 1$

C

D

X

9 root.T:

$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = (b_n)^n$

① $a_n = ()^n$

② $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} a_n$

< 1

> 1 or ∞

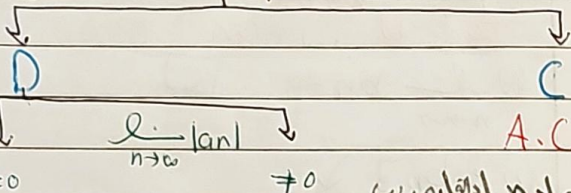
$= 1$

C

D

X

⑩ if $\sum |a_n|$



① $= 0$
 ② $f(x) < 0$
 C.C

⑪ L.C. + $\frac{0}{0}$

حد درجه $\sin()$
 حد درجه e^n
 حد درجه

- ① $b_n =$ استخراج
- ② $\sum b_n =$ استخراج قواسم و ثابت ها
- ③ $\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty}$

$\therefore$ basis by L.C.

⑪ Taylor. S

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

- ① $t = x - a \Rightarrow x = t + a$
- ② $f(x) =$ اعوض x ب $t+a$

$\therefore f(x) =$ قواسم

⑩ powe. S

- ① $a_n =$, $a_{n+1} =$
- ② using root or ratio
- ③ $\lim_{n \rightarrow \infty}$

④ $< X <$
 at $x =$ at $x =$

اعوض x ب $t+a$ و a ب t
 استخدم a, p, q

⑤ interval is $[c]$
 $x \in (a, b)$

$$\sum_{n=0}^{\infty} a_n (x-c)^n$$

⑫ MacLaurin. S

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

- ① $t = x - a \Rightarrow x = t + a$
- ② $f(x) =$

$\therefore f(x) =$ قواسم